

CRMMS – 24-Month Study:

Before First Pass:

- CBRFC sends out the Lake Powell Unregulated Inflow Forecast
 - LC sends out message page and forecast data to LC interested parties.
 - UC begins their pass back process among themselves
 - Green->Aspinall->San Juan->Glen Canyon (generally)
- Meeting with Water Accounting
 - Receive MRIDs for any updating accounting forecasts along with any changes to system conservation and other water use assumptions.
 - After this meeting, and before beginning the modeling process, we will receive a summary of changes from Water Accounting that then gets located in COM460\24MONTH\YYYY\MM - MMMYY\QAQC_SupportingDocs\Annotated Agenda for Coord Call w RiverOps YYYY-DD-YY.docx
- Meeting with MWD
 - LC presents most recent forecast with MWD; MWD provides us with projected monthly diversions (1-2 month outlook) and projected yearly total diversion.

First Pass

Initialization:

- Upon email receipt from the Glen Canyon Operator (c. Alex P.), LC begins first pass process.
- The following files are used during the the
 - 24MS Input: COM460\24MONTH\YYYY\MM - MMMYY\QAQC_SupportingDocs\24MS Input_MMMYY.xlsm
 - Updating various information, either manually or with macros, such as:
 - End of month elevation targets for Mohave/Havasui
 - MWD/CAP diversion targets
 - Hoover 17-month schedule
 - Binary Hoover Operations Projections (BHOPs)
 - Outage schedule for Davis/Parker

- **Manual Schedule Adjustments – MMMYY.xlsm** -
COM460\24MONTH\YYYY\MM - MMMYY\QAQC_SupportingDocs\ Manual
Schedule Adjustments – MMMYY.xlsm
 - Process varies a bit at this point depending on MX and PVID orders/changes.
 - Adjustments to BHOPS w/ instructions listed in the sheet
 - Keep book open past this point for three states' checks, to take place later in the first pass process.
- COM460\24MONTH\YYYY\MM -
MMMYY\QAQC_SupportingDocs\ **Projected State Use-MMMYY.xlsx**
 - Documents water use assumptions for individual LB states
 - Details system conservation, frequently/semi-frequently adjusted
 - Used after initialization process, to be detailed more later.
- Opening current month CRMMS Model – starting with the MOST probable
 - Initial name will be along the lines of CRMMS_vX.YY.mdl; create copy model file renamed-> CRMMS_vX.YY_MMMYY_MOST.mdl (or .mdl.gz)
 - Run script 2: “Set Up 24MS Run First Pass”
 - Begin by setting the run range, Start date in the current month, end date *may* vary depending on time of year. Generally, it will be through September of outyear 2.
 - For November and December, the finish range will go to the end of 24 months following the current month. (Ex: Nov25 – Oct 27; Dec25- Nov27).
 - This will take you through the login of UCHDB, enter your credentials, you will then want to pick which model via the model table - this will be Model ID 2 “AOP Model”
 - After that is selected, you will look for the current month model run ID (MRID). As of March 2026, the Most Probable MRID is 3308 and will be noted in the Powell operator’s email to the 24MS coordination group.
 - For UCHDB, you will have to select this SDI 3 times, it will be the same MRID each time.
 - After this, you will enter your credentials for LCHDB.
 - First you will be prompted to enter the Model Run ID for LC Current Operations (noted in bold) – this will still use Model ID 2 “AOP Model.”
 - IMPORTANT: You will not want to use the same MRID here as you did for the UC selections. Instead, you

will want to use the MRID from the 24-Month Study Most Probable from the previous month.

- This will be selected twice.
- Next, will be asked to select the Model Run ID for “Official Water Use Forecast Current Year,” “Official Water Use Forecast Outyear 1,” and “Official Water Use Forecast Outyear 2”
 - Select Model ID 27 - “Water Use Forecast Model” then you can select the MRIDs that Water Accounting has provided from the most recent coordination meeting.
 - The script will then run the model after all data has been pulled in from HDB.
- Once the First Pass script has been completed, you will want to open the 24MS Input Sheet as mentioned before. This spreadsheet has several tabs, starting with Input
 - Input –
 - Select the month in a dropdown menu
 - Update the MWD diversion for the current and upcoming month
 - Depending on time of year and any circumstances, update the Davis/Parker maintenance schedules
 - This takes place on separate tabs

Month	Days in Month	Davis Unit 1		
		Date Off	Last Day Off or Last Day of Month	Total Days On
Jan-26	31			31
Feb-26	28			28
Mar-26	31			31
Apr-26	30			30
May-26	31			31
Jun-26	30			30
Jul-26	31			31
Aug-26	31			31
Sep-26	30			30
Oct-26	31	20	31	19
Nov-26	30	1	26	4
Dec-26	31			31

Figure 1 Sample of Davis/Parker maintenance spreadsheet.

- Figure 1 shows that the date a unit comes off and a date it comes back online are both recorded, along with a separate column for the number of days a unit is online.
- If necessary, update the Lake Mohave and Lake Havasu end of month elevation targets
- Finally, you can use the buttons for “Get CAP Data,” “Get Hoover 17-Month Schedule,” and “Get Havasu Release,” to finish with updates. The CAP and 17-month schedule will have you navigate to separate folders to select a file.
 - The CAP folder path will be:
N:\D4000\COM460\24MONTH\YYYY\CAP Schedules
 - The 17-month schedule will be:
N:\D4000\COM460\24MONTH\YYYY\Hoover 17-Month Schedule
- The next and final step with the Input spreadsheet is to copy the columns from the output tab into the Input tab of the CRMMS Monthly SCT, as shown in figures 2 and 3.

DATE	AZ CU Schd CAP (ac-ft)	MWDDiv Sched w/o TJ (ac-ft)	Havasu Outflow BHOPSParker (af/month)	Mohave Target Contents (kaf)	Havasu Target Contents (kaf)	EnergyInfo Effective Hoover (MW)	Hoover Maint Factor NONE	Davis Maint Factor NONE	Parker Maint Factor NONE
Mar-26	158 120	65,000	606 391	1745 18	521 94	1446	0.4703	0.7871	0.9597
Apr-26	182 320	89,000	670 423	1759 29	521 94	1404	0.5345	0.8000	1.0000
May-26	172 940					1399	0.9180	0.8000	1.0000
Jun-26	21 400					1387	0.9732	0.9933	1.0000
Jul-26	12 054					1399	0.9732	1.0000	1.0000
Aug-26	15 560					1390	0.9748	1.0000	1.0000
Sep-26	74 500					1404	0.8479	1.0000	1.0000
Oct-26	54 000					1404	0.5837	0.8903	0.7823
Nov-26	41 000					1419	0.5783	0.6267	0.7867
Dec-26	30 000					1432	0.9200	0.6065	0.8548
Jan-27						1446	0.7884	0.6129	0.7742
Feb-27						1446	0.6134	0.6143	0.7679
Mar-27						1432	0.6980	0.7613	0.9032
Apr-27						1404	0.8853	0.9800	1.0000
May-27						1374	0.9733	1.0000	1.0000
Jun-27						1348	1.0000	1.0000	1.0000
Jul-27						1323	1.0000	1.0000	1.0000
Aug-27								1.0000	1.0000
Sep-27								1.0000	1.0000
Oct-27								0.8903	0.7500
Nov-27								0.6267	0.7867
Dec-27								0.6065	0.9113
Jan-28								0.6129	0.7742
Feb-28								0.6143	0.7679
Mar-28								0.7613	0.9032
Apr-28								0.9800	1.0000
May-28								1.0000	1.0000
Jun-28								1.0000	1.0000

Copy this
entire
selection for
the 24 MS
Monthly SCT.

Figure 2 Columns as shown in the 24MS Input spreadsheet

Import paste the data starting in the CAP slot at the first timestep.

Timestamp	Day	Arizona_OU_Schedule CAP acre-ft	MRIIDConversion Sched wired T3 acre-ft	Havasu Outflow acre-ft/month	HavasuOutflow ParkerScheduleRequirement acre-ft/month	HavasuOutflow BHOPSParkerOutflow acre-ft/month	Mohave Target Contents 1,000 acre-ft	Havasu Target Contents 1,000 acre-ft	EnergyInfo EffectivePowerCapacityInput MW	EnergyInfo PercentInflowInAvailableInput NONE	Davis Maint Factor NONE	Lower Basin Parker Plan NONE
12/31/25	Wed	45,758	42,300	215,021	NaN	176,312.00	NaN	NaN	1,418.00	0.49	0.61	
1/1/26	Sat	70,344	26,869	245,739	NaN	247,724.00	NaN	NaN	1,460.00	0.58	0.75	
1/2/26	Sun	102,911	0	253,786	NaN	358,078.00	NaN	NaN	1,474.00	0.58	0.60	
1/3/26	Mon	122,824	86,000	315,047	615,947	615,947.00	1,746.18	521.94	1,448.00	0.47	0.60	
1/4/26	Tue	182,320	63,922	680,968	680,213	680,968.00	1,759.29	544.61	1,418.00	0.53	0.60	
1/5/26	Sun	172,940	83,879	719,096	719,095	719,096.00	1,759.29	544.61	1,390.00	0.62	0.61	
1/6/26	Tue	21,400	96,451	668,038	668,038	668,038.00	1,759.29	544.61	1,399.00	0.97	0.99	
1/7/26	Fri	12,054	96,000	639,318	639,318	639,318.00	1,731.11	531.38	1,399.00	0.97	1.00	
1/8/26	Mon	15,560	96,000	517,775	517,775	517,775.00	1,731.11	521.94	1,404.00	0.97	1.00	
1/9/26	Wed	74,500	96,000	447,224	447,224	447,224.00	1,675.37	521.94	1,404.00	0.97	1.00	
1/10/26	Sat	54,000	75,000	459,625	Set By Rule: (185) Set Parker Schedule Requirement to 459,625	459,625.00	1,487.05	521.94	1,404.00	0.84	0.89	
1/11/26	Mon	41,000	93,069	339,698	345,267	339,698.00	1,539.75	521.94	1,404.00	0.84	0.63	
1/12/26	Thu	30,000	93,069	308,615	312,937	NaN	1,725.50	503.41	1,432.00	0.84	0.61	
1/13/26	Mon	29,877	19,414	353,879	400,894	NaN	1,725.50	503.41	1,432.00	0.92	0.61	
1/14/26	Wed	246,844	55,433	614,236	646,935	NaN	1,759.29	507.09	1,432.00	0.85	0.76	
1/16/26	Fri	280,615	66,930	712,864	716,537	NaN	1,759.29	544.61	1,390.00	0.85	0.98	
1/17/26	Mon	284,042	85,986	783,089	785,237	NaN	1,759.29	544.61	1,274.00	0.95	1.00	
1/18/26	Wed	113,066	83,493	735,887	735,400	NaN	1,759.29	544.61	1,335.00	1.00	1.00	
1/19/26	Fri	18,989	85,312	689,477	696,384	NaN	1,731.11	531.38	1,295.08	0.97	1.00	
1/20/26	Sun	31,181	81,311	597,179	604,489	NaN	1,731.11	521.94	NaN	0.97	1.00	
1/21/26	Tue	82,384	74,585	516,590	533,779	NaN	1,675.37	521.94	NaN	0.97	1.00	
1/22/26	Thu	88,426	77,149	521,216	526,324	NaN	1,487.05	521.94	NaN	0.84	0.89	
1/23/26	Sun	77,149	80,208	388,642	395,124	NaN	1,539.75	521.94	NaN	0.84	0.63	
1/24/26	Mon	62,561	79,709	308,113	314,327	NaN	1,661.57	503.41	NaN	0.46	0.61	
1/25/26	Wed	97,159	63,486	295,279	312,935	NaN	1,725.50	503.41	NaN	0.78	0.61	
1/26/26	Fri	93,394	19,414	385,143	400,847	NaN	1,725.50	503.41	NaN	0.85	0.61	
1/27/26	Sun	258,352	55,433	625,479	646,935	NaN	1,759.29	507.09	NaN	0.79	0.76	
1/28/26	Mon	293,688	66,930	704,346	716,535	NaN	1,759.29	544.61	NaN	0.79	0.98	
1/29/26	Wed	292,284	85,986	774,745	785,235	NaN	1,759.29	544.61	NaN	0.95	1.00	
1/30/26	Fri	123,456	83,493	717,502	722,452	NaN	1,759.29	544.61	NaN	0.97	1.00	
1/31/26	Mon	19,466	85,312	680,505	696,402	NaN	1,731.11	531.38	NaN	0.97	1.00	
2/1/26	Tue	32,614	81,311	597,162	604,489	NaN	1,731.11	521.94	NaN	0.97	1.00	
2/2/26	Thu	61,354	74,585	517,547	533,779	NaN	1,675.37	521.94	NaN	0.97	1.00	
2/3/26	Sat	92,549	77,149	NaN	NaN	NaN	1,487.05	521.94	NaN	0.84	0.89	
2/4/26	Sun	75,303	80,208	NaN	NaN	NaN	1,539.75	521.94	NaN	0.84	0.63	
2/5/26	Mon	44,962	79,709	NaN	NaN	NaN	1,661.57	503.41	NaN	0.46	0.61	

Figure 3 SCT cells in which the 24MS input data are to be copied into.

After Initialization, Manual Schedule Adjustments:

- Start with the Manual Schedule Adjustments, as mentioned earlier, and go to the “Start” tab. You will once again need to enter in the MRIDs for Water Use Forecasts that Water Accounting provides.

Current Year MRID	Out Year 1 MRID	Out Year 2 MRID
1001558	1001489	1001493

Figure 4 Except for January model runs, you will need three MRIDs

- After entering the MRIDs, you can select each of the buttons to the side of these yellow highlighted cells:
 - Update AZ Historical & Current Year Schedule
 - Update AZ Outyear Schedules
 - Update CA Historical & Current Year Schedule
 - Update CA Outyear Schedules
 - BHOPS
 - Update Mexico
- There is a final button, “Reset Excel Sheet,” that is only used at the start of a new calendar year.
- Typically, we proceed to the “California Users” tab and look at any necessary adjustments for PVID.
 - These adjustments may vary year by year due to any ongoing agreements for following that are in place. This part of the process can sometimes be dependent on any additional information we receive from Water Accounting/Water Contracts.

		Actual				
Compare HDB to annual approval		Month/Year	HDB Schedule	Monthly Adj	Adj Schedule	Distribution of HDB Sched
2026 Estimate	387072	1/31/2026	7442		7442	2.4%
HDB Sched	306042	2/28/2026	18018		18018	5.9%
Annual Difference	81030	3/31/2026	23603		23603	7.7%
Full Following	0	4/30/2026	32612		32612	10.7%
FollowingFactor	4.48	5/31/2026	41918		41918	13.7%
Conserved Amount	0	6/30/2026	44323		44323	14.5%
		7/31/2026	45116		45116	14.7%
Monthly Adjustment		8/31/2026	40614		40614	13.3%
Number of Months to Adj		9/30/2026	26311		26311	8.6%
		10/31/2026	15386		15386	5.0%
Sum - Adj Sched	306042	11/30/2026	8021		8021	2.6%
		12/31/2026	2678		2678	0.9%
2027 Estimate	420696	1/31/2027	3219	-32	3187	0.7%
HDB Sched	449286	2/28/2027	22001	-2360	19641	4.9%
Annual Difference	-28590	3/31/2027	39807	-4749	35058	8.9%
Full Following (acres)	25,947	4/30/2027	45343	-6331	39012	10.1%
Following Factor Adjustment	0.50	5/31/2027	64528	-8255	56273	14.4%
Following Conservation Factor (af/ac)	4.5	6/30/2027	69122	-8978	60144	15.4%
Conserved Amount	58,381	7/31/2027	58998	-9121	49877	13.1%
Monthly Adjustment		8/31/2027	66109	-8635	57474	14.7%
Reduction Jan-Jul @50%	34055	9/30/2027	47646	-6060	41586	10.6%
Reduction Aug-Dec @50%	24325	10/31/2027	26332	-2915	23417	5.9%
Sum - Adj Sched	390905	11/30/2027	3282	-906	2376	0.7%
		12/31/2027	2899	-38	2861	0.6%
2028 Estimate	368464	1/31/2028	3219	-32	3187	0.7%
HDB Sched	449286	2/29/2028	22001	-2360	19641	4.9%
Annual Difference	-80822	3/31/2028	39807	-4749	35058	8.9%
Full Following	25,947	4/30/2028	45343	-6331	39012	10.1%
Following Factor Adjustment	0.50	5/31/2028	64528	-8255	56273	14.4%
Following Conservation Factor (af/ac)	4.5	6/30/2028	69122	-8978	60144	15.4%
Conserved Amount	58,381	7/31/2028	58998	-9121	49877	13.1%
Monthly Adjustment		8/31/2028	66109	-8635	57474	14.7%
Reduction Jan-Jul @50%	34055	9/30/2028	47646	-6060	41586	10.6%
Reduction Aug-Dec @50%	24325	10/31/2028	26332	-2915	23417	5.9%
Sum - Adj Sched	390905	11/30/2028	3282	-906	2376	0.7%
		12/31/2028	2899	-38	2861	0.6%
Notes: 4.48 Following Factor was based on the 2010 Verification Report found at: 2010 Verification Report						

Figure 5 The schedules for outyear 1 and outyear 2 will receive adjustments based on expected following agreements to be executed.

- Verify the Mexico schedules – this may require opening the most recent MX schedules file, but there should be a total delivery that initially adds up to 1.5 maf OR 1.45 maf (due to shortage)
 - These values of 1.5 or 1.45 are then reduced by MX water reserve creation, recoverable water through the Binational Water Scarcity Contingency Plan (BNWSCP), system conservation, and water reserve delivery (figure 3).

1	2	3	4	5	6	7	8
Month	Total Delivery at NIB	Tijuana Delivery	Mexico Water Reserve Creation	Mexico Recoverable	Min 330 System Conservation	Mexico Water Reserve Delivery	Total Delivery (sum of 2,4,5,6)
1/31/2026	252,803	0	-126,392	-30,000	0	0	96,411
2/28/2026	116,225	0	-11,297	0	0	0	104,928
3/31/2026	142,010	0	0	0	0	1,541	143,551
4/30/2026	140,371	0	0	0	0	4,203	144,574
5/31/2026	109,439	0	0	0	0	4,343	113,782
6/30/2026	119,777	0	0	0	0	4,203	123,980
7/31/2026	119,861	0	0	0	0	4,343	124,204
8/31/2026	103,142	0	0	0	0	3,014	106,156
9/30/2026	95,612	0	0	0	0	0	95,612
10/31/2026	69,485	0	0	0	0	0	69,485
11/30/2026	93,494	0	0	0	0	0	93,494
12/31/2026	87,783	0	0	0	0	0	87,783
TOTAL (US Units)	1,450,000	0	-137,689	-30,000	0	21,647	1,303,958

Figure 6 MX Water table from Manual Schedule Adjustments

- Next, you will want to go to the BHOPS release Changes tab, in which there is a box with instructions on what you will need to do, reading:
 - Need to copy and paste **Havasu outflow.parker schedule requirement** from model into column F. Button queries historical data and daily model CFS from Havasu (see data to right).
- At this point, you can minimize the Manual Schedule adjustments. It may be needed later for the Three states check while making adjustments to MWD or CAP.

Adjusting Yearly apportionments; Projected State Use:

During this part, you will want to have the Projected State Use file open, which can be found in the QAQC folder like many of the other files mentioned. Other than in January, this spreadsheet will have tabs for the current calendar year and two outyears in which we will be completing most of our accounting of our water use assumptions.

The general process here will be repeated for each of the Most Probable, Probable Min, and Probable Max. During this, you will need to be aware of the most current assumptions for MWD and SNWA annual diversions; MWD is provided during the monthly conference call, SNWA is provided via the more recent quarterly jp water use assumptions call.

Each LB state will have slightly different processes, to be outlined below.

California –

- Primarily determined via MWD planned diversion and CA ics banks.
- The state apportionment in the RiverWare model is updated with the Projected State Use value
 - This value can change monthly for various reasons including:
 - Historic water use

- Changes in water schedules
- Changes in system conservation
- The California state apportionment may need to be “tinkered” with in order to get the MWD annual use correct (as the ICS banks permit)
 - This requires changes to MWD ICS creation/delivery (creation = - number, delivery = +number)

Arizona –

- CAP/CAWCD’s annual forecasted use is updated differently than MWD. CAPs forecasted use is also reported/modeled differently depending on if we’re in current year or outyears.

Arizona	Values in AF	Notes
Basic apportionment	2,800,000	
Shortage volume (IF YEAR 1)	0	
CAP - Storage of NV unused	0	CAP annual diversion of 0.862 maf
AZ - ICS (delivery or creation)	0	
Estimated payback(s)	0	
PSCP (BHC 2)	-2,477	
AZ - System Water	-257,725	See System Conservation tab
Estimated AZ adjusted apportionment for Model	2,539,798	
DCP Contribution	-192,000	192 kaf DCP contribution with 27479 af DCP ICS
CAWCD ICS creation/delivery	0	
Shortage Volume if Outyear	-320,000	shortage volume
Modeled AZ state total for year 1	2,027,798	
Unused - interstate banking	0	
Unused - left in system	0	
ICS converted to system water due to full bank	0	DCP ICS converted to system water due to full bank

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2026

2027

2028

System Conservation

SysCon SummaryTable

DCP Bank